

WILKES COUNTY AP, NC, USA

WMO: 723165

Lat: 36.223N

Lon: 81.098W

Elev: 396

StdP: 96.66

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-6.0	-19.2	0.7	-1.6	-17.2	0.9	-1.3	8.9	4.2	8.0	4.6	1.7	310	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.0	23.0	32.4	22.6	31.2	22.1	25.1	30.4	24.3	29.3	23.7	28.4	2.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.4	27.2	22.8	18.4	26.1	22.4	17.9	25.8	78.4	30.3	75.5	28.6	72.7	28.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	5.8	5.0	-12.0	34.8	3.4	1.9	-14.4	36.2	-16.4	37.4	-18.3	38.5	-20.8	39.9		
			-12.7	25.6	3.2	1.1	-15.0	26.4	-16.9	27.1	-18.7	27.7	-21.0	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	3.8	5.4	9.5	14.4	18.8	23.0	24.5	23.8	20.6	14.9	9.2	5.6
	DBStd	8.37	5.38	4.73	5.09	4.37	3.59	2.67	2.27	2.67	3.15	4.24	4.61	4.37
	HDD10.0	656	201	142	73	12	0	0	0	0	0	8	69	151
	HDD18.3	2069	449	363	275	129	39	2	0	1	16	123	275	396
	CDD10.0	2300	10	13	58	143	274	389	449	429	318	158	45	13
	CDD18.3	671	0	0	2	11	54	141	191	171	84	15	1	0
	CDH23.3	5794	1	1	33	173	504	1246	1632	1411	633	142	16	1
	CDH26.7	2112	0	0	3	33	131	484	673	569	191	27	1	0
Wind	WSAvg	1.4	1.7	1.8	2.0	2.0	1.4	1.2	0.9	0.9	1.1	1.2	1.4	1.3
Precipitation	PrecAvg	1248	109	80	124	102	97	117	117	123	107	91	79	102
	PrecMax	1611	219	168	218	212	248	210	178	271	258	292	182	210
	PrecMin	931	29	30	37	21	20	36	75	30	9	2	19	33
	PrecStd	186	51	36	48	58	51	51	30	63	55	61	41	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	22.2	27.1	29.8	32.2	35.1	34.9	36.1	32.5	29.2	26.0	22.2
		MCWB	13.5	15.7	16.4	17.5	20.7	22.9	24.2	23.2	20.6	20.5	16.1	15.3
	2%	DB	17.6	18.8	23.6	27.4	29.9	32.8	32.9	32.9	31.0	27.1	22.4	17.6
		MCWB	11.6	12.0	14.7	16.4	20.2	22.5	23.4	23.1	20.8	18.5	14.1	12.9
	5%	DB	15.2	17.0	21.3	25.9	27.7	31.3	32.2	32.0	28.8	24.8	20.1	15.2
		MCWB	10.7	10.6	13.1	16.0	19.3	21.8	23.1	22.6	20.6	17.0	13.9	11.8
	10%	DB	12.6	13.8	18.0	22.8	26.3	29.8	30.9	30.0	27.3	22.5	17.6	12.8
		MCWB	8.4	7.7	11.5	14.7	18.7	21.5	22.7	22.6	20.2	16.6	12.2	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	17.1	17.9	20.1	22.9	24.7	26.2	26.2	24.0	22.9	18.4	17.4
		MCDB	18.3	19.4	24.1	24.2	28.2	31.3	31.7	31.7	27.8	25.9	21.0	18.0
	2%	WB	13.7	14.7	16.1	18.5	21.7	23.9	25.2	25.1	23.0	21.2	16.9	14.9
		MCDB	16.9	17.6	20.7	24.5	26.5	30.0	30.5	30.6	27.1	23.4	19.9	16.9
	5%	WB	11.4	11.6	14.9	17.4	20.8	23.1	24.4	24.1	22.2	19.6	15.3	12.4
		MCDB	13.9	14.8	19.4	22.0	25.6	28.8	29.5	28.7	26.1	23.1	18.6	14.5
	10%	WB	9.0	8.9	13.0	16.2	19.9	22.4	23.8	23.5	21.3	17.8	13.0	10.1
		MCDB	12.3	13.4	17.0	20.8	24.8	27.5	28.5	27.7	24.9	20.6	16.8	12.4
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.1	11.9	12.8	12.2	12.2	11.3	11.2	11.2	12.3	12.7	10.2
		MCDBR	14.5	15.4	15.7	16.6	14.6	14.8	13.5	13.9	14.5	15.3	16.2	13.6
		MCWBR	9.6	9.8	8.3	7.7	6.6	5.8	5.2	5.1	5.8	7.5	9.1	9.0
	5% WB	MCDBR	11.9	13.2	13.1	13.2	12.2	13.0	11.7	11.4	11.3	11.4	11.9	11.1
	MCWBR	9.2	9.9	8.2	7.2	6.0	5.7	5.0	5.0	5.5	6.5	8.5	8.6	
Clear-Sky Solar Irradiance	taub	0.305	0.314	0.343	0.393	0.436	0.471	0.511	0.509	0.403	0.358	0.317	0.302	
	taud	2.499	2.467	2.426	2.293	2.193	2.133	2.047	2.031	2.320	2.434	2.513	2.537	
	Ebn at Noon	896	928	926	890	852	820	783	775	850	864	874	877	
	Edh at Noon	81	94	108	130	146	155	168	167	119	97	80	74	
All-Sky Solar Radiation	RadAvg	2.46	3.10	4.18	5.22	5.74	6.16	5.73	5.29	4.50	3.65	2.77	2.11	
	RadStd	0.17	0.36	0.27	0.41	0.45	0.44	0.40	0.27	0.41	0.45	0.24	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+150	+91

Nomenclature: See separate page